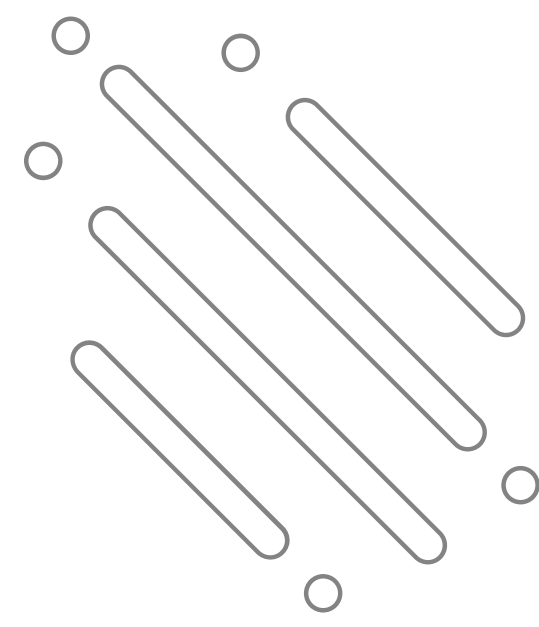
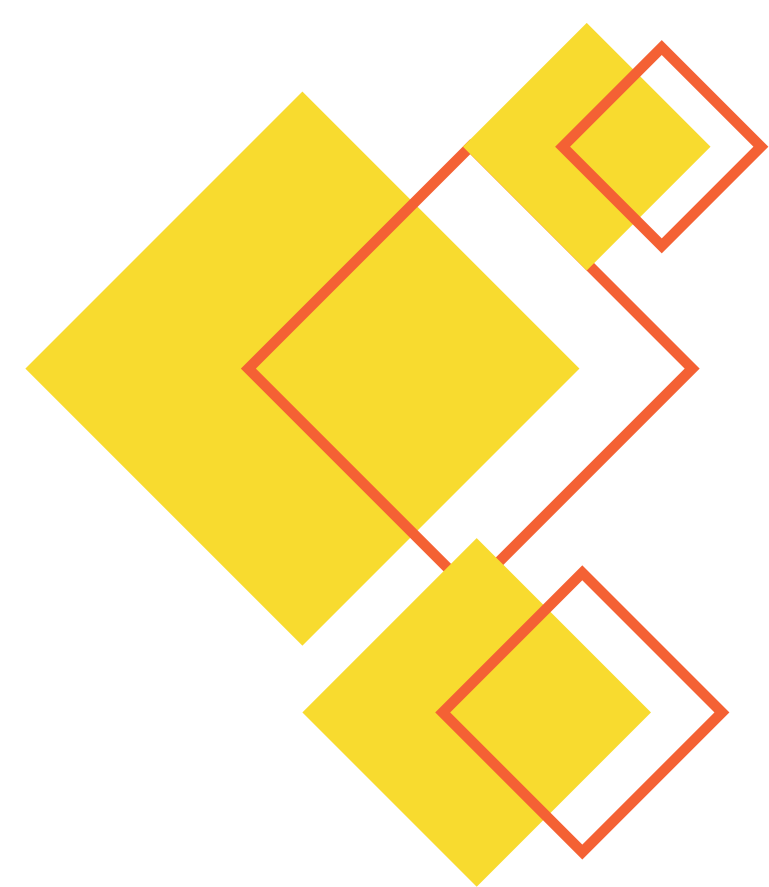
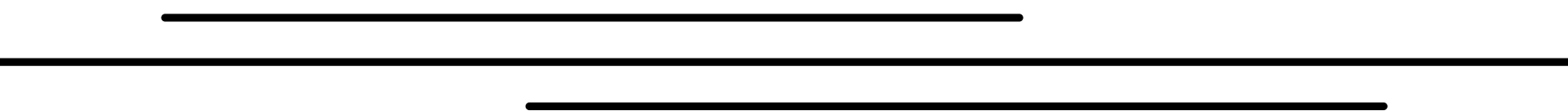




FIELDGUARD

AI-Powered Offline Attendance Verification for NHAI
Highway Workers



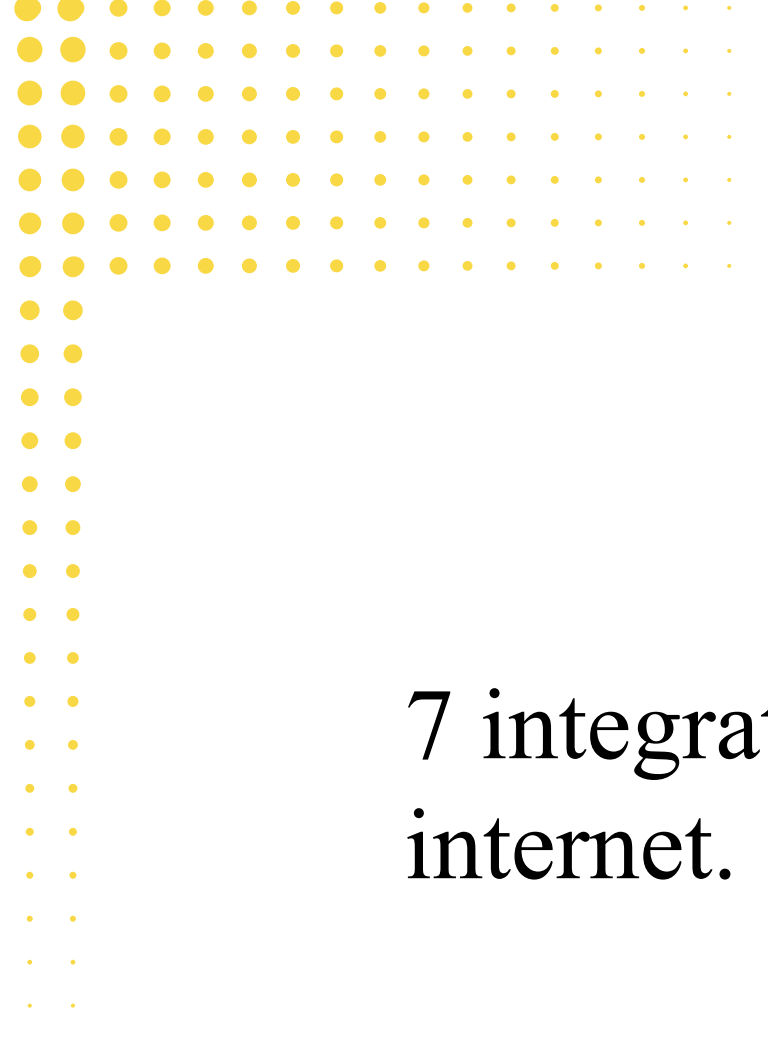
THE PROBLEM

Consider, Ravi is a highway worker 200 km from the city. No internet. No signal. Middle of nowhere. His company needs to know: Did Ravi actually show up today?

1. **Proxy attendance** : A colleague clocks in using someone else's ID card or phone
2. **Ghost workers** : Workers marked present at sites they never visited
3. **No connectivity** : Remote NHAI sites have zero or unreliable internet
4. **Paper registers** : Manual registers are easy to forge and hard to audit

Current solutions fail because they require internet, trust paper, or are easily spoofed.

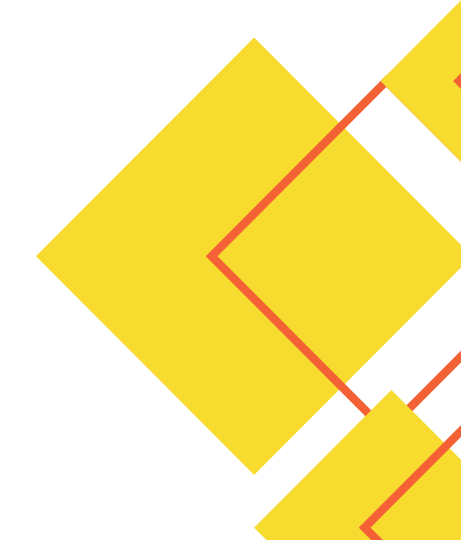


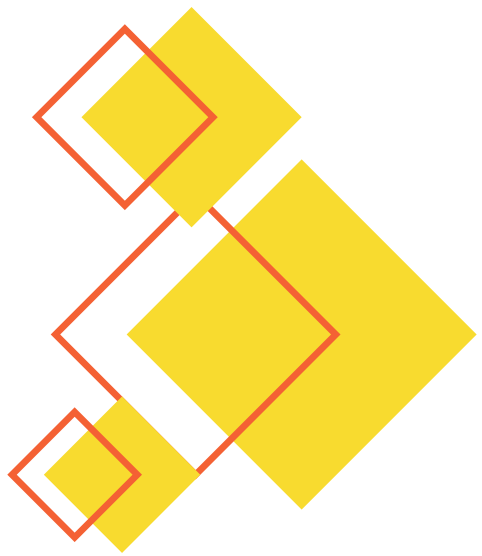


SOLUTION OVERVIEW

7 integrated modules working fully offline by syncing when connected to internet.

Flow:

1. Face Detection
 2. Liveness Detection
 3. GPS + Sensor Fusion
 4. Presence Pulse
 5. Risk Engine
 6. Offline Merkle Storage
 7. Delta Sync to AWS → Supervisor Dashboard
- 



FACE RECOGNITION + LIVENESS DETECTION

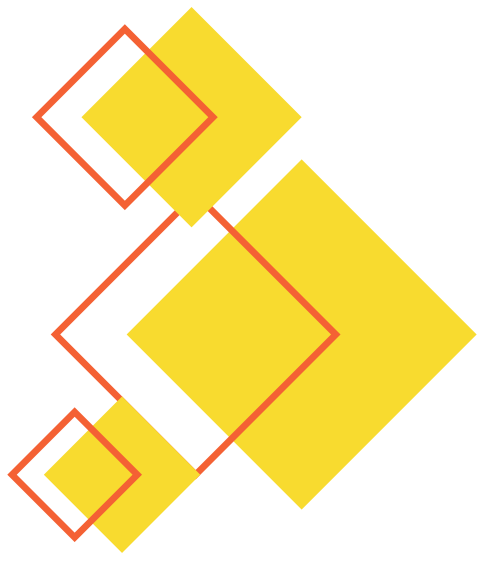
Module 1 : Face Detection & Recognition

Stage 1 - BlazeFace (0.6 MB) :

- Detects face in frame in ~30ms. Rejects blank frames immediately - MobileFaceNet never wastes cycles on non-face frames.

Stage 2 - MobileFaceNet (5.1 MB) :

- Runs only on confirmed face crops. Generates 128-dim embeddings matched against locally stored (AES-encrypted) worker embeddings.
- Enrollment mode (pre-deployment) : Supervisor captures 5–7 frames per worker on WiFi. Embeddings stored encrypted in local SQLite.
- Indian demographic fix Model fine-tuned on IIT Indian face dataset. Skin tone normalization at preprocessing handles harsh sunlight & shadow.



TWO-LAYER LIVENESS DETECTION

Layer A :

Passive CNN

Runs on every frame with no user action. Detects moiré patterns, specular reflections, and micro-texture gradients to distinguish real skin from printed photos or screens.

Layer B :

Active challenge :

The challenge based verification is shown only when passive score is 0.4–0.7.

Random challenges are generated like blink, turn left, right, nod.



GPS + PRESENCE VERIFICATION

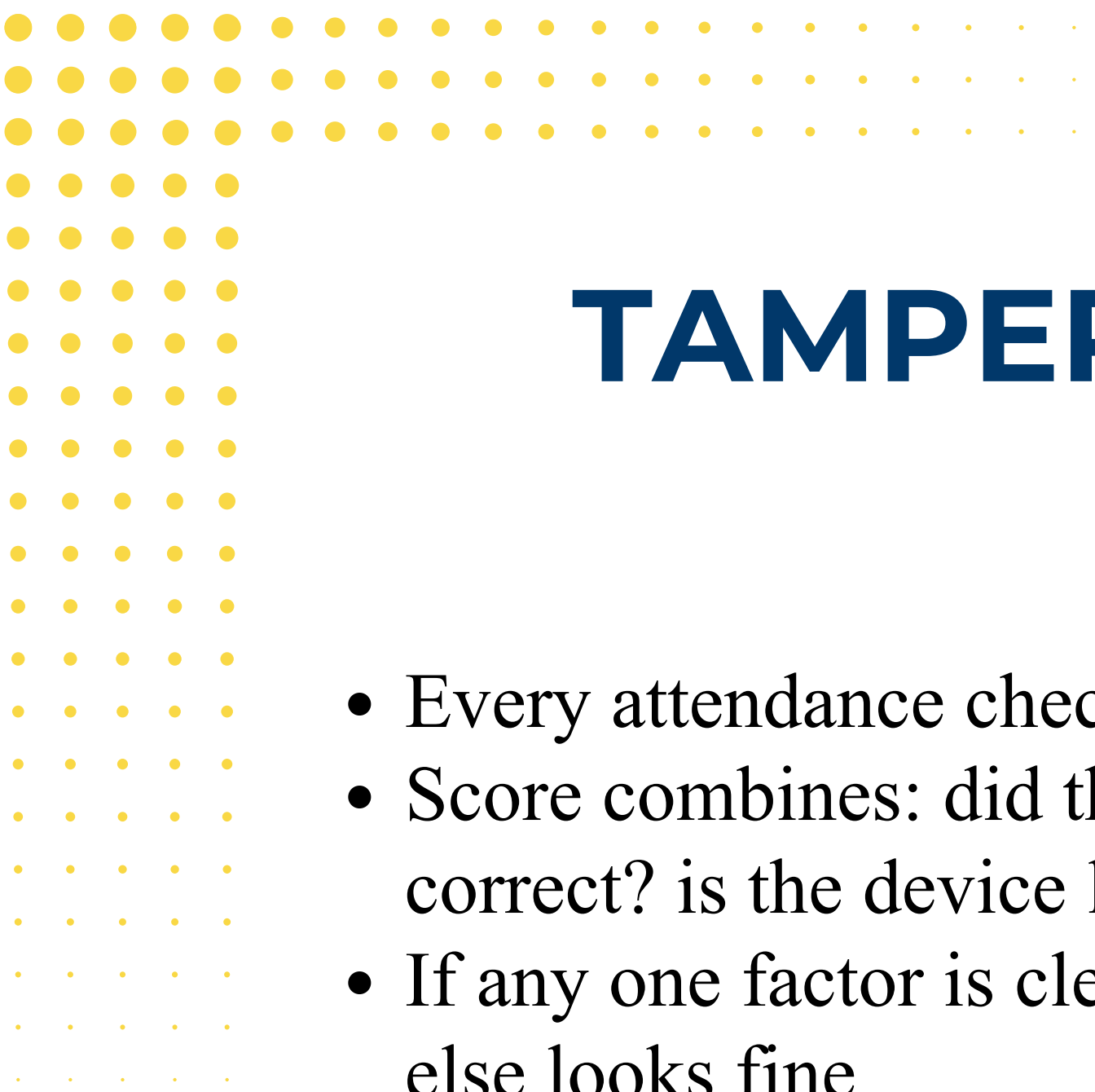
- App confirms the worker is physically standing at the correct highway site - not marking attendance from home or a nearby town
- Three separate checks happen at the same time - if at least two of them agree, the worker is marked present
- Random check-ins happen throughout the shift - workers never know when the next one will come, so they can't just show up for the morning and leave

Check	What it does	How we do it
GPS	Checks the phone's coordinates against the site's saved location	Phone's built-in GPS - site coordinates saved inside the app before deployment
Nearby WiFi BSSID Scans	Scans for WiFi hotspots near the site (construction vehicles, site office routers) to confirm proximity within ~50m of site equipment.	Phone's WiFi scanner - completely free, needs no internet connection
Phone movement	Confirms the worker is standing upright and moving normally - not sitting in a car far away from the worksite	Phone's built-in accelerometer and barometer (same sensors used for step counting)

BAYESIAN RISK ENGINE

Formula : $P(\text{fraud}) = 1 - P(\text{face}) \times P(\text{liveness}|\text{face}) \times P(\text{zone}) \times P(\text{device}) \times P(\text{temporal})$

- Every attendance check gets a trust score - not just a pass/fail
- Score combines: did the face match? was it a real person? was the location correct? is the device legitimate? is the timing normal?
- If any one factor is clearly wrong, the entire score drops - even if everything else looks fine
- Score slowly decreases if the worker stops responding to checks during the day
- Over time, the system learns each worker's normal pattern and flags unusual behaviour



TAMPER-EVIDENT OFFLINE STORAGE

- Every attendance check gets a trust score - not just a pass/fail
- Score combines: did the face match? was it a real person? was the location correct? is the device legitimate? is the timing normal?
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- Over time, the system learns each worker's normal pattern and flags unusual behaviour



SECURE STORAGE & CLOUD SYNC

- All attendance data is locked with a key that only that specific phone can access
- Even if someone steals the phone and extracts the data, they cannot read or edit it
- Every record is linked to the one before it — if anyone tampers with old records, it is immediately detected
- When internet is available, only new unsynced records are uploaded — works even on very slow 2G connections
- If the upload fails halfway, nothing is lost — the app tries again until it gets confirmation

TECH STACK

App framework
React Native 0.73+

Face detection
BlazeFace TFLite · 0.6 MB

Face recognition
MobileFaceNet INT8 · 5.1 MB

Active liveness
MediaPipe FaceMesh · 2.8 MB

Passive liveness
Silent-Face-Anti-Spoofing · ~3 MB

ML runtime
react-native-fast-tflite (NNAPI)

Local DB
SQLite (expo-sqlite) · WAL + AES-256-GCM

Key storage
Android Keystore / iOS Secure Enclave

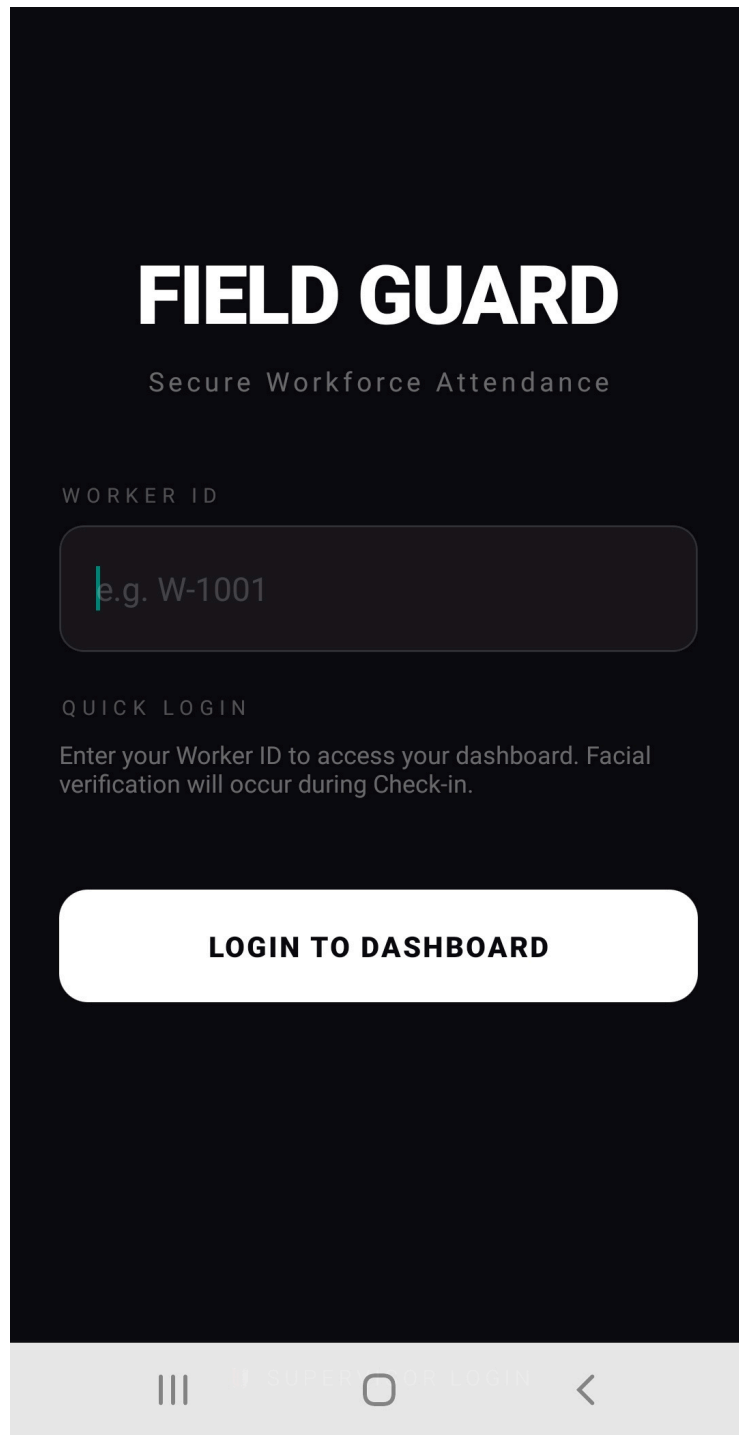
Location
expo-location + WiFi BSSID scan

Backend
AWS Lambda + DynamoDB + API Gateway

Alerts
AWS SNS push notifications

Integrity
SHA-256 Merkle hash chain

IMPLEMENTATION RESULTS



FIELD GUARD
Secure Workforce Attendance

WORKER ID

e.g. W-1001

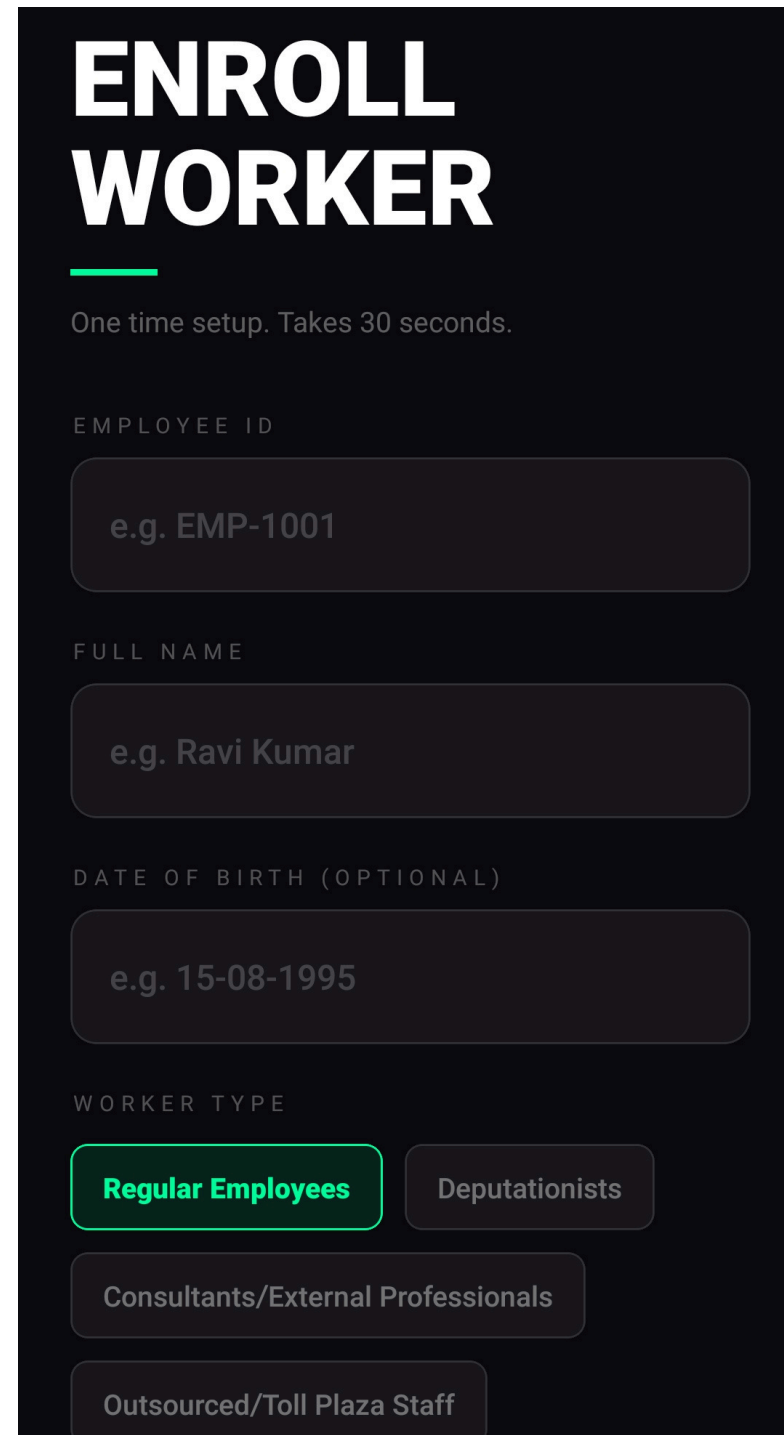
QUICK LOGIN

Enter your Worker ID to access your dashboard. Facial verification will occur during Check-in.

LOGIN TO DASHBOARD

Navigation bar: Home, Profile, Settings, Logout

THE FIELD GUARD APPLICATION



ENROLL WORKER

One time setup. Takes 30 seconds.

EMPLOYEE ID

e.g. EMP-1001

FULL NAME

e.g. Ravi Kumar

DATE OF BIRTH (OPTIONAL)

e.g. 15-08-1995

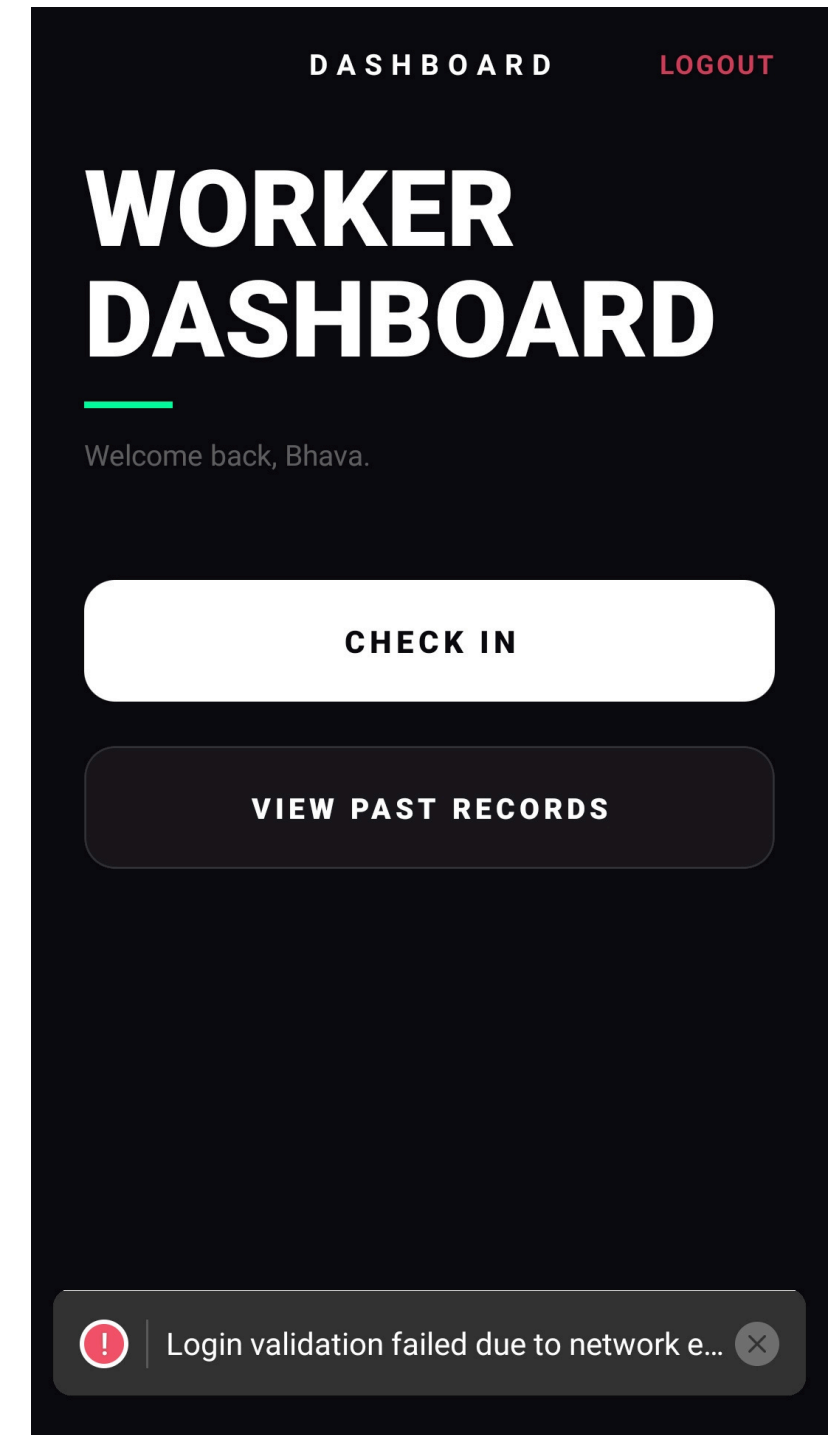
WORKER TYPE

Regular Employees Deputationists

Consultants/External Professionals

Outsourced/Toll Plaza Staff

ENROLLING A NEW USER



DASHBOARD **LOGOUT**

WORKER DASHBOARD

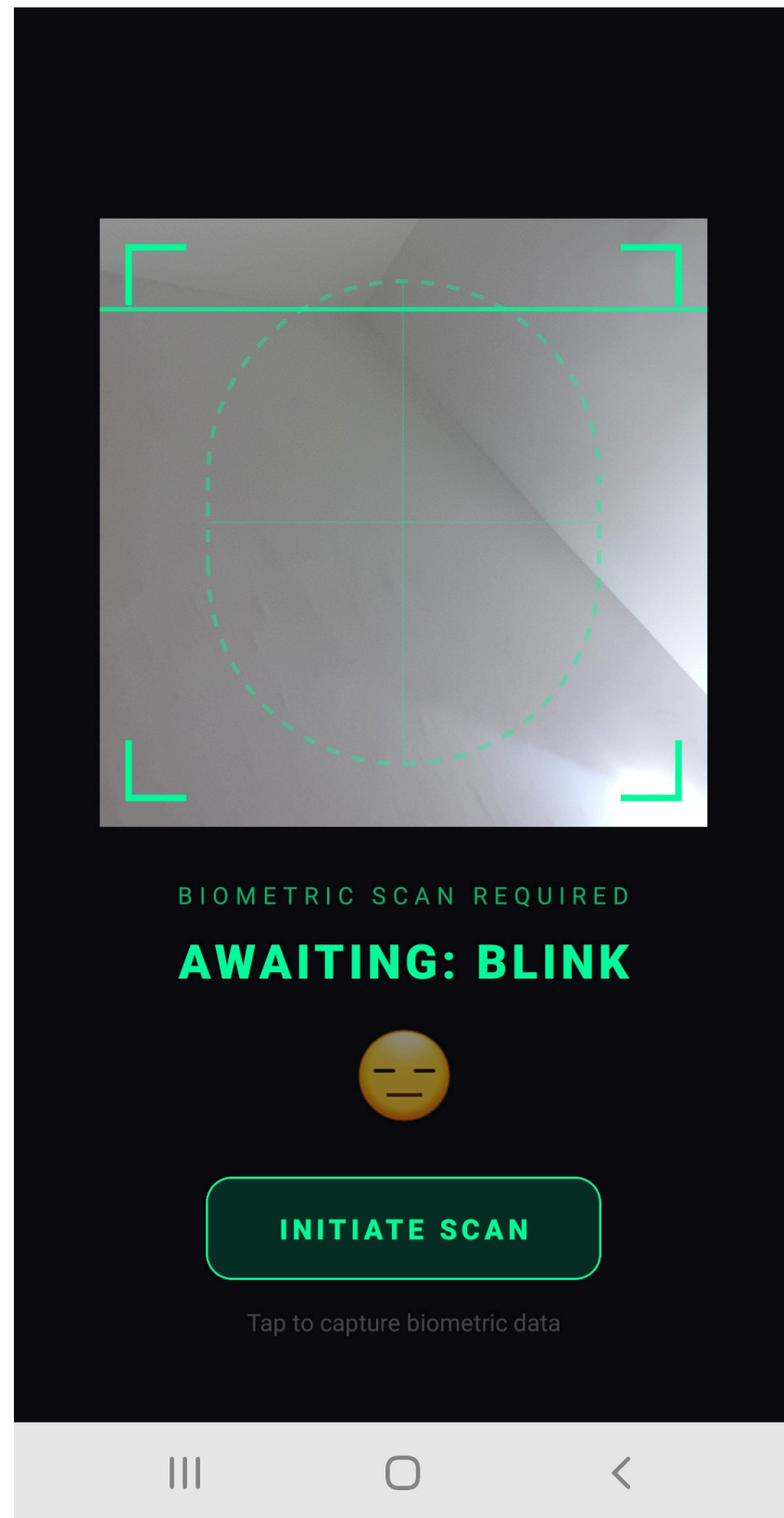
Welcome back, Bhava.

CHECK IN

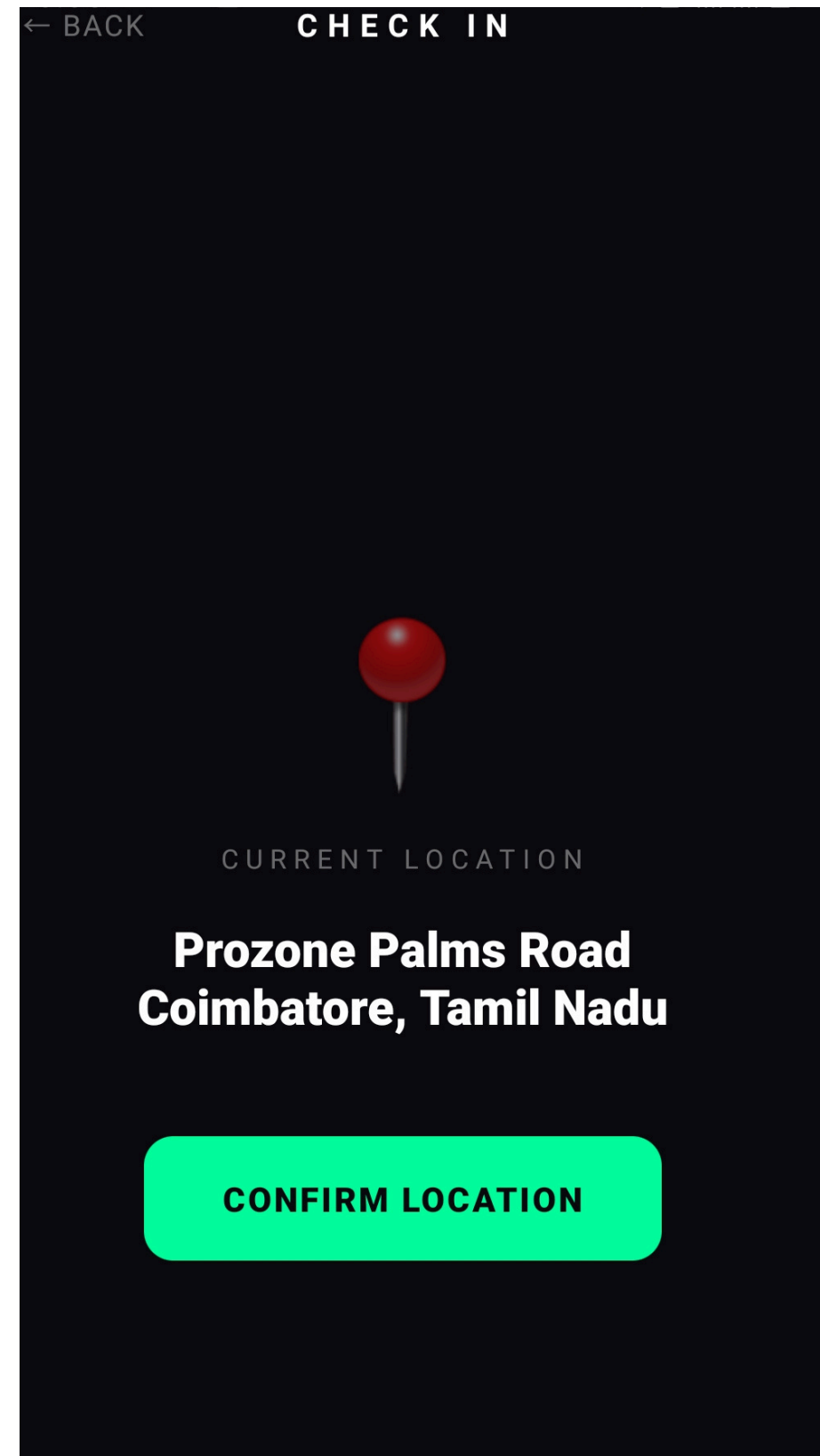
VIEW PAST RECORDS

! Login validation failed due to network e... ✕

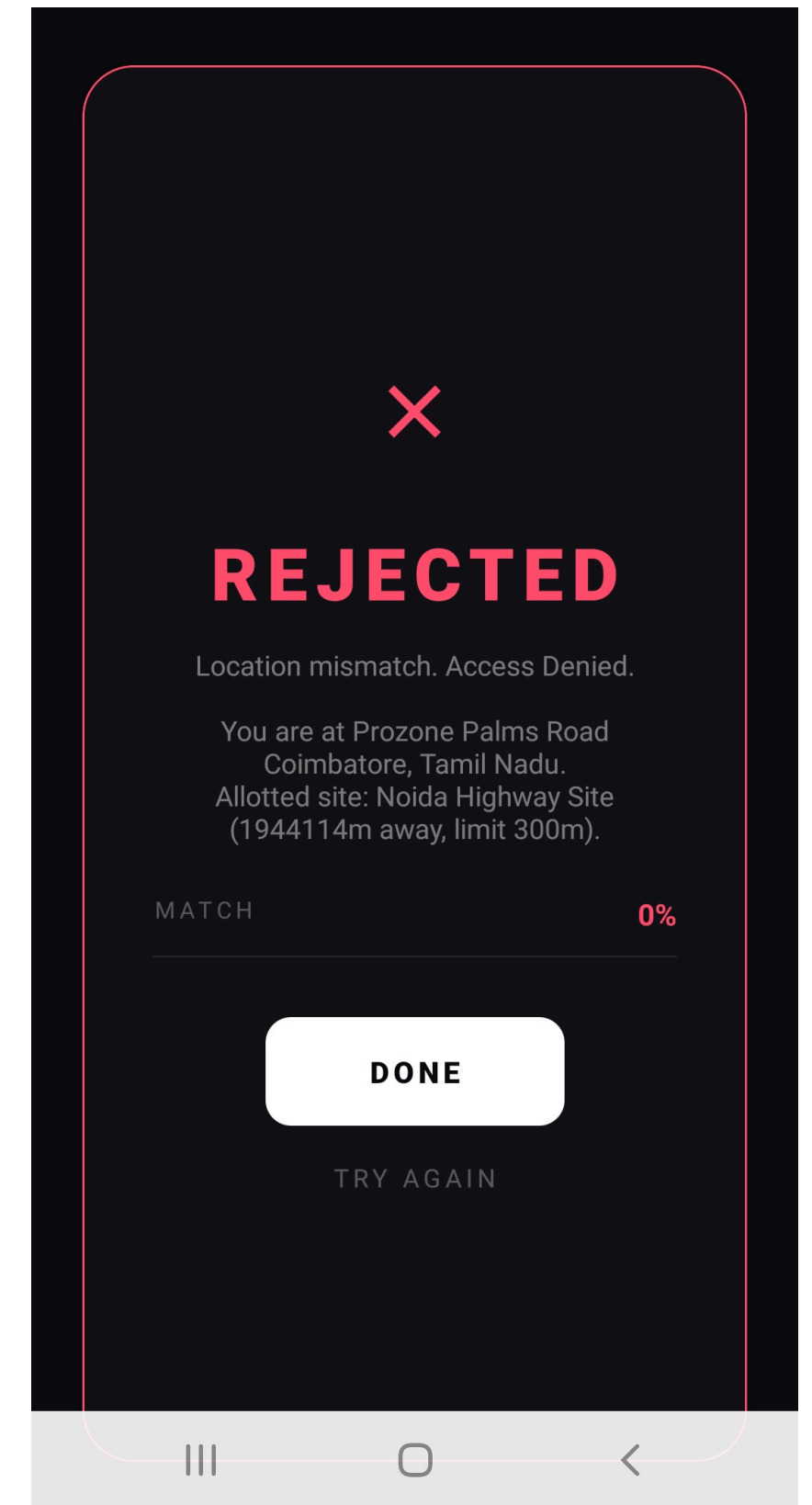
WORKER DASHBOARD



FACE DETECTION + LIVENESS DETECTION



VERIFYING WORKER'S LOCATION



ATTENDANCE REJECTED WHEN LOCATION MISMATCHED



THANK YOU

